Assignment 2: Polynomial Interpolation

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1. Assignment Statement

Select two functions and calculate their interpolation polynomials using the Lagrange, Newton, and Hermite methods. Compare the original functions with the interpolation polynomials by increasing the number of interpolation points.

2. Source Code

The Python implementation of this assignment is available on GitHub.

Repository link: https://github.com/HananeINHID/Fondements-mathematiques-IA.git

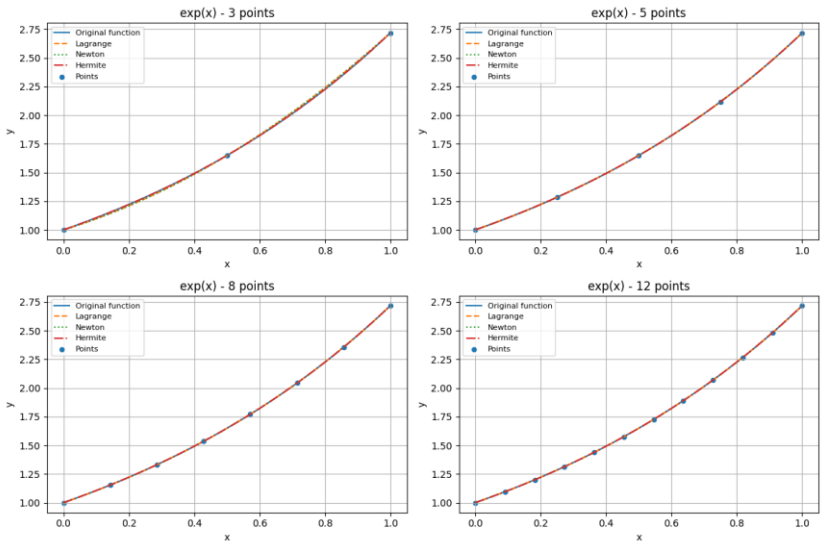
File: assignment2.py

3. Results and Interpretation

3.1. First Function: $f(x) = e^x$

Results:

```
Function: exp(x)
------------------------------------------------------------------------
Points | Lagrange Error | Newton Error | Hermite Error
------------------------------------------------------------------------
     3 |    1.442065e-02 |  1.442065e-02 |  5.577193e-06
     5 |    5.265688e-05 |  5.265688e-05 |  5.963674e-12
     8 |    4.807574e-09 |  4.807575e-09 |  9.769963e-15
    12 |    2.153833e-14 |  6.217249e-15 |  1.241673e-12
```



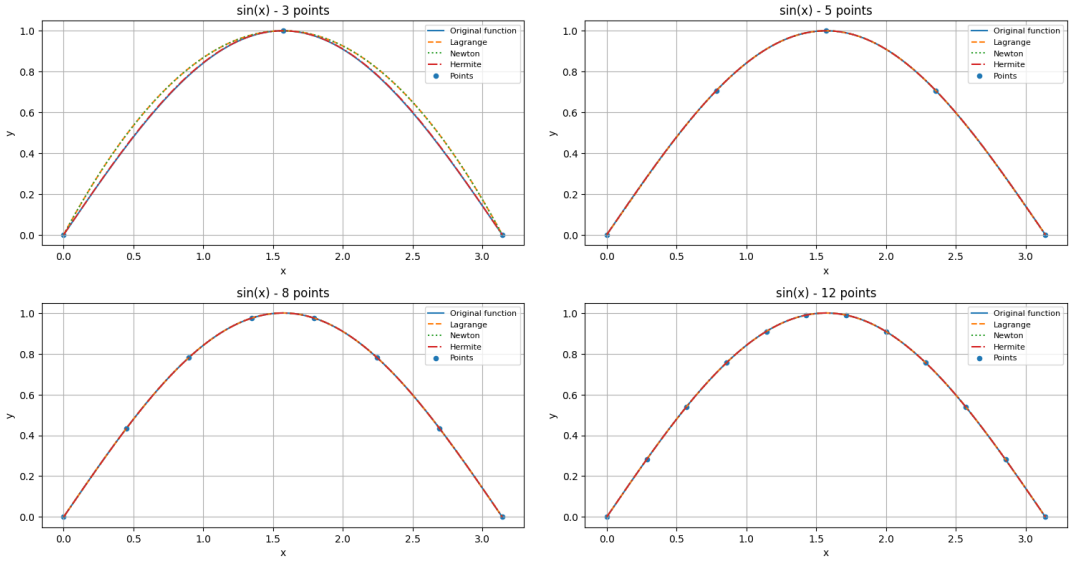
Interpretation:

The graphs show that the three methods give results close to the original function. When the number of interpolation points increases, the error decreases. Hermite gives smaller errors than Lagrange and Newton in these tests because it uses both function values and derivative values.

3.2. Second Function: $f(x) = \sin(x)$

Results:

```
Function: sin(x)
--------------------------------------------------------------------------------
Points | Lagrange Error | Newton Error | Hermite Error
--------------------------------------------------------------------------------
     3 |    5.600959e-02 |  5.600959e-02 |  2.787061e-03
     5 |    1.811962e-03 |  1.811962e-03 |  3.057607e-07
     8 |    2.438074e-05 |  2.438074e-05 |  5.673240e-14
    12 |    2.625259e-09 |  2.625263e-09 |  7.362166e-13
```



Interpretation:

For the sine function, we observe the same result. With only 3 points, the difference between the original function and the interpolation polynomials is more visible. When we increase the number of points to 5, 8, and 12, the approximation becomes more accurate. Hermite makes smaller errors in our experiments.

3.3. Comparison

Lagrange and Newton give almost the same results because they construct the same polynomial interpolation using the same points. Hermite uses the derivatives as additional information, which explains its higher accuracy in these examples.

4. Conclusion

This assignment helped us understand and compare the three interpolation methods. The results show that increasing the number of points improves the approximation for both functions. Hermite also makes very small errors because it takes the derivatives into account